

# Higher-Order Structure in Residue-Class Transitions of Consecutive Primes

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<https://github.com/thinkthoughts/prime-numbers-lab>

## Abstract

We encode consecutive primes as residue classes modulo 30, producing a sequence over 8 states  $\{1, 7, 11, 13, 17, 19, 23, 29\}$ . We construct a first-order transition operator  $P$  and compare the empirical two-step operator  $P^{(2)}$  to the baseline  $P^2$  via  $\Delta = P^{(2)} - P^2$ .

Using the first  $N$  primes,  $\Delta$  has low-rank structure. Low-rank approximations improve prediction of  $P^{(2)}$  relative to  $P^2$ .

Synthetic controls produce distinct residual operators. Residue-class transitions of consecutive primes modulo 30 are represented by the residual operator  $\Delta = P^{(2)} - P^2$ , which exhibits higher-order structure.

## 1 Introduction

We study transitions of consecutive primes. Reducing them modulo 30 yields a sequence over 8 residue states  $\{1, 7, 11, 13, 17, 19, 23, 29\}$ .

We construct a transition operator  $P$  and compare  $P^{(2)}$  to  $P^2$ . The residual

$$\Delta = P^{(2)} - P^2$$

isolates second-order structure.

We test whether  $\Delta$  reflects random variation or systematic structure.

## 2 Related Work

Classical models of prime gaps include probabilistic heuristics such as Cramér’s model [3] and analytic frameworks for prime distributions [4].

Markov models provide a baseline for transition structure [1], while information-theoretic measures quantify higher-order structure relative to first-order statistics [2].

This work analyzes residue-class transitions of consecutive primes using empirical transition operators and the residual operator  $\Delta = P^{(2)} - P^2$ .

## 3 Transition Framework

Let  $\{x_n\}_{n=1}^N$  be residues modulo 30 of the first  $N$  primes.

We estimate

$$P_{ij} = \frac{\#\{n : x_n = i, x_{n+1} = j\}}{\#\{n : x_n = i\}}.$$

Define

$$\Delta = P^{(2)} - P^2.$$

## 4 Results

Figure 1 shows the transition operator  $P$ .

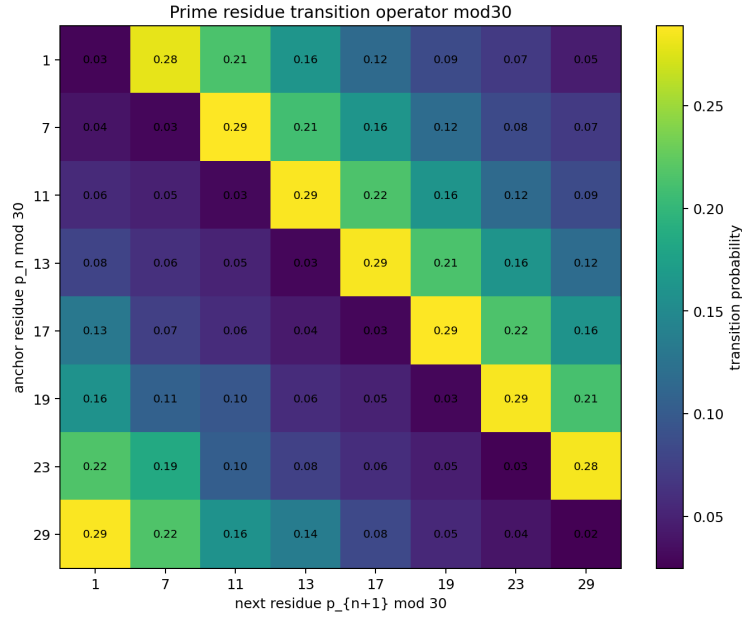


Figure 1: First-order transition operator  $P$ .

Figure 2 shows the residual  $\Delta = P^{(2)} - P^2$ , with structured deviations.

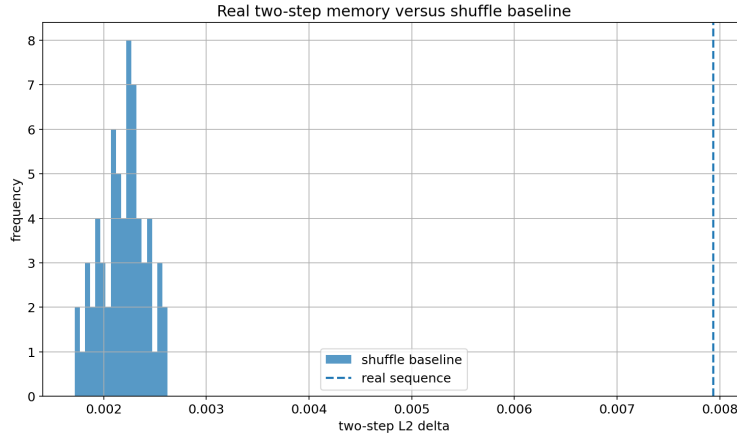


Figure 2: Residual operator  $\Delta = P^{(2)} - P^2$ .

Figure 3 shows low-rank structure in  $\Delta$ .

Figure 4 shows statistical validation across controls.

Figure 5 shows summary ratios for empirical and control statistics.

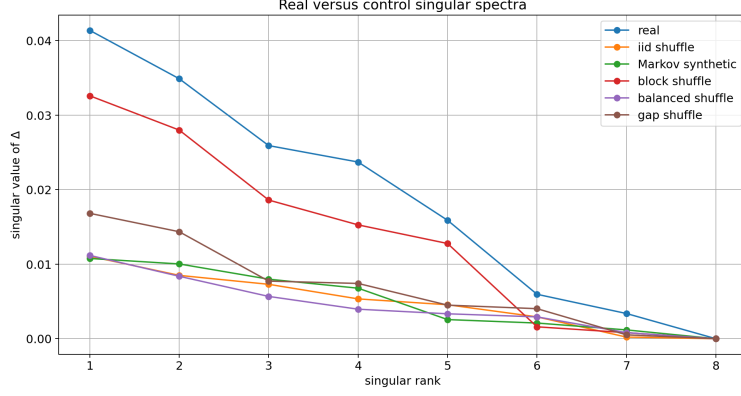


Figure 3: Low-rank structure of  $\Delta$ .

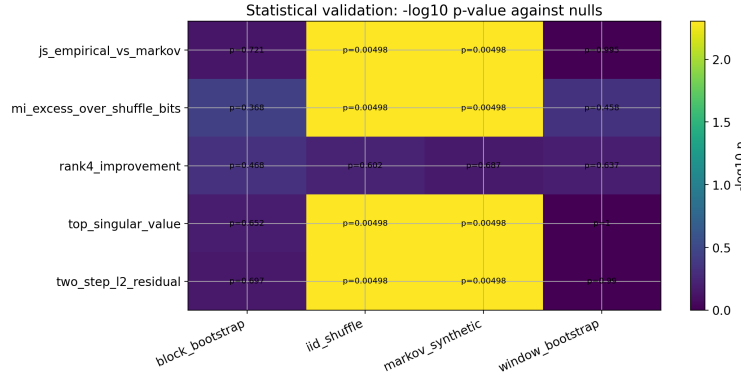


Figure 4: Statistical validation p-values.

## 5 Validation

The prime sequence differs from iid and Markov control sequences in the residual operator  $\Delta$ . Block resampling shows partial agreement, indicating local structure.

## 6 Discussion

The operator  $\Delta$  isolates second-order structure relative to  $P$ . Its structure is low-rank, with deviations concentrated in a small number of components.

A small number of components approximates  $P^{(2)}$  more accurately than  $P^2$ .

This structure persists across sample sizes  $N$ .

## 7 Conclusion

Residue-class transitions of consecutive primes modulo 30 are represented by the residual operator  $\Delta = P^{(2)} - P^2$ , which exhibits higher-order structure.

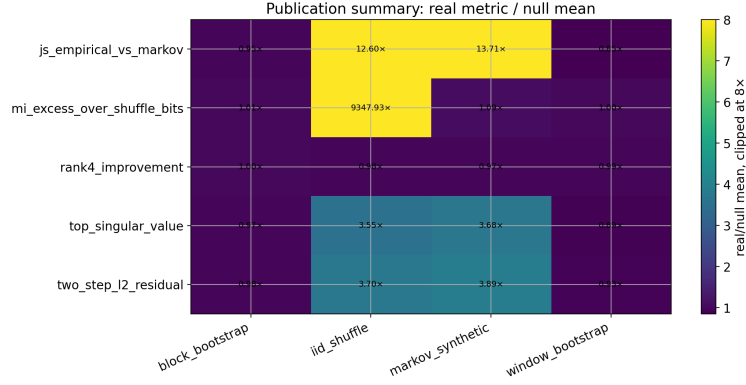


Figure 5: Summary ratios for empirical and control statistics.

## References

- [1] David Aldous. Random walks on finite groups and rapidly mixing markov chains. *Seminaire de Probabilites*, 1983.
- [2] Thomas Cover and Joy Thomas. *Elements of Information Theory*. Wiley, 2006.
- [3] Harald Cramér. On the order of magnitude of prime gaps. *Acta Arithmetica*, 1936.
- [4] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*. Oxford University Press, 1979.